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11.4.4.3 sin() function

The **sin()** function takes a single argument in radians. The **sin()** function returns the value in the range of [-1, 1]. The returned value is either in double, float, or long double.

11.4.4.4 pow() function

The **pow()** function returns base raised to the power of exponent. If any argument passed to **pow()** is long double, the return type is promoted to long double. If not, the return type is double. The **pow()** function takes two arguments:

- **base** - the base value
- **exponent** - exponent of the base

Program 11.14

```
#include <iostream>
#include <math.h>
using namespace std;
int main ()
{
        double base, exponent, result;
        base = 5;
        exponent = 4;
        result = pow(base, exponent);
        cout << "pow("<<base << "^" << exponent << ") = " << result;
        double x = 25;;
        result = sin(x);
        cout << "\nsin("<<x<<")= "<<result;
        return 0;
}
```

Output:

```
pow(5^4) = 625
sin(25)= -0.132352
```

11.4.5 Generating Random Numbers

The **srand()** function in C++ seeds the pseudo random number generator used by the **rand()** function. The seed for **rand()** function is 1 by default. It means that if no **srand()** is called before **rand()**, the **rand()** function behaves as if it was seeded with **srand(1)**. The **srand()** function takes an unsigned integer as its parameter which is used as seed by the **rand()** function. It is defined in<**cstdlib**>or <**stdlib.h**>header file.